

7.6 (a) Using the boundary condition, it is straightforward to show that

$$E_i + E_r = E_t, \quad E_i - E_r = n(\omega) E_t,$$

which lead to,

$$E_r = \frac{1-n(\omega)}{1+n(\omega)} E_i, \quad E_t = \frac{2}{1+n(\omega)} E_i.$$

Since the reflected wave is in the same medium as the incident wave, the reflection coefficient

is simply $R = \left| \frac{E_r}{E_i} \right|^2 = \left| \frac{1-n(\omega)}{1+n(\omega)} \right|^2$

and the transmission coefficient can be determined by energy conservation

$$T = 1 - R = \frac{4 \operatorname{Re}[n(\omega)]}{|1+n(\omega)|^2}$$

(b) The transmitted wave has the full spatial dependence as

$$\vec{E}(x, y, z, t) = \vec{e} E_i \exp\left\{i \frac{n(\omega)\omega}{c} z - i\omega t\right\},$$

where $\vec{e}$ is polarization vector. Then, $\vec{D}(x, y, z, t) = \vec{e} \epsilon E_i \exp\left\{i \frac{n(\omega)\omega}{c} z - i\omega t\right\},$

and $\operatorname{Re}\left[i\omega \vec{E} \cdot \vec{D}^*/2\right] = \frac{1}{2} \omega \operatorname{Im}[\epsilon(\omega)] \exp\left\{-2 \operatorname{Im}[n(\omega)] \omega z/c\right\}$